TCP Socket Programming

1. What is TCP?

TCP (Transmission Control Protocol) is a connection-oriented transport layer protocol. It establishes a connection between the client and server before data is exchanged.

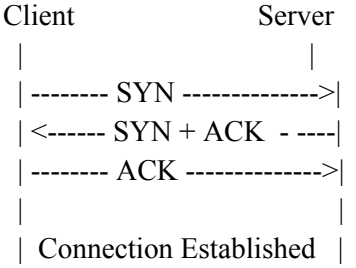
TCP provides:

- Reliable delivery of data
- Ordered delivery of data
- Retransmission of lost data
- Error detection
- Connection-oriented communication

Unlike UDP, TCP ensures that data sent by the application is delivered reliably and in the correct order.

2. TCP Three-Way Handshake

Before exchanging application data, the TCP client and server establish a connection using a **three-way handshake**:

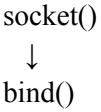


After the handshake, the client and server can exchange application data.

3. TCP Client-Server Communication

A typical TCP client-server program follows these steps.

Server

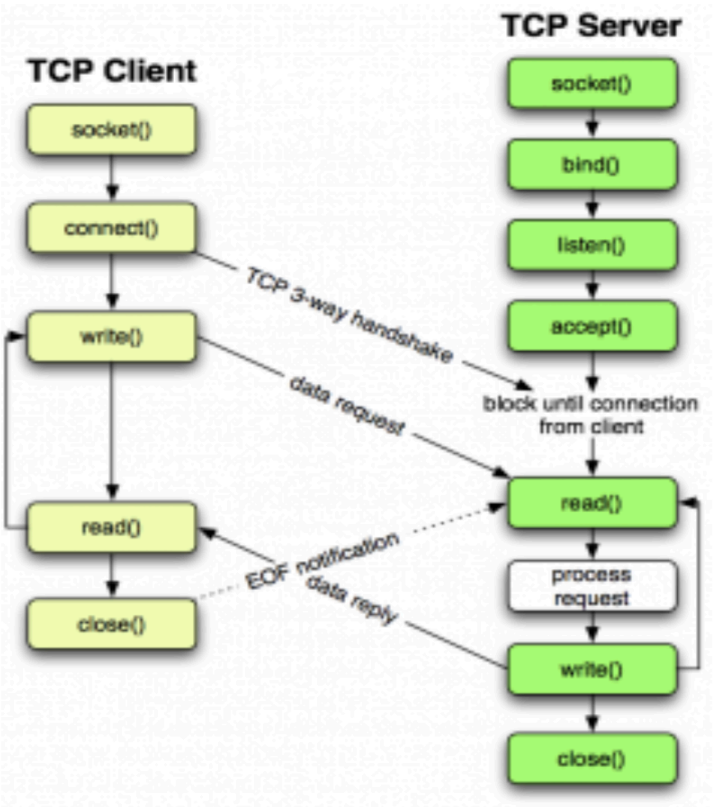


↓
listen()
↓
accept()
↓
recv() / send()
↓
close()

Client

socket()
↓
connect()
↓
send() / recv()
↓
close()

Complete Communication Flow



Important Socket Functions

Function	Purpose
socket()	Creates a TCP socket
bind()	Assigns an IP address and port to the server
listen()	Makes the server ready to accept connections
accept()	Accepts a connection from a client
connect()	Connects the client to the server
send()	Sends data through the TCP connection
recv()	Receives data from the TCP connection
close()	Closes the socket

Note: The server uses accept() to establish a connection with a client. After that, communication takes place using send() and recv().

4. TCP vs UDP

Both TCP and UDP are transport layer protocols, but they work differently.

TCP	UDP
Connection-oriented	Connectionless
Reliable delivery	No delivery guarantee
Ordered data delivery	No ordering guarantee
Retransmits lost data	Does not retransmit lost data
Uses connection establishment	No connection establishment
Generally higher overhead	Lower overhead

Why do we need TCP if UDP already exists?

UDP is useful when speed and low overhead are more important than guaranteed delivery. However, applications such as file transfer and many client-server applications require data to arrive **reliably and in the correct order**. TCP handles this reliability automatically.